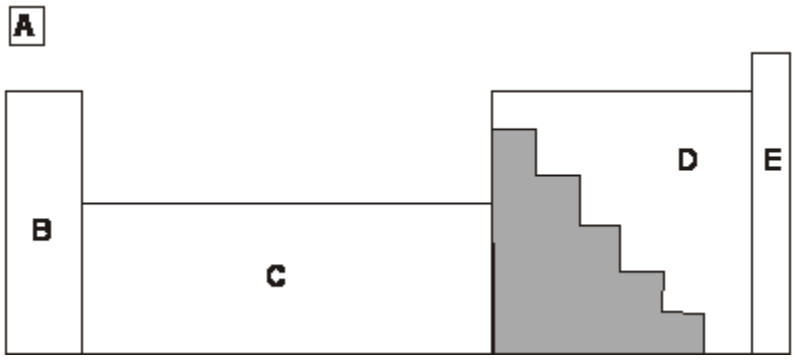
Q1.

(a) The diagram below shows part of the periodic table of elements.



The shaded area contains **only** metal elements.

Two other areas also contain **only** metal elements.

Which areas contain only metal elements?
Tick the **two** correct boxes.

A ☐ B ☐ C ☐ D ☐ E ☐

1 mark

(b) Copper is a metal.

At room temperature copper is a strong solid.
Give **two** other properties of copper that show it is a metal.

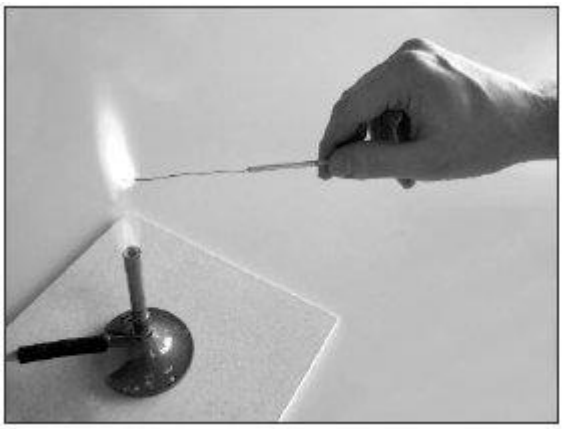
1. .....

1 mark

2. .....

1 mark

(c) When copper metal is heated it reacts with a gas in air.



What is the chemical name of the **product** formed when copper reacts with a gas in air?

.....

1 mark

(d) Which statement below describes what happens in a **chemical change** but **not** in a physical change?

Tick the correct box.

The product is a solid. ☐

The change only happens at a high temperature. ☐

The atoms have combined in a different way to make a new substance. ☐

The types of atoms at the start are the same as in the end product. ☐

1 mark

maximum 5 marks

Q1.

(a) • B ✓
C ✓

__both__ answers are required for the mark
if more than two boxes are ticked, award no mark

1 (L6)

(b) any **two** from

- it conducts electricity
- it conducts heat

one mark may be awarded for 'it is a good conductor' if a reference to heat or electricity is not given

- it is ductile or malleable

'it bends' is insufficient

- it has a high melting **or** boiling point

accept 'it is shiny'
accept 'it is sonorous'
accept 'it forms basic oxides'
'it is strong or hard' is insufficient

2 (L5)

(c) copper oxide

accept 'CuO'
do __not__ accept 'copper dioxide'

1 (L6)

(d) The atoms have combined in a different way to make a new substance. ✓

if more than __one__ box is ticked, award no mark

1 (L6)

[5]